MEMORANDUM
OFFICE OF THE STATE ENGINEER
Hydrology Bureau

DATE: April 30, 2003

TO: Paul Saavedra, Chief, Water Rights Division

FROM: Andy Core, Hydrology Bureau

CC: Tom Morrison, Chief, Hydrology Bureau OK TL

SUBJECT: Residual Effects of 1972-1987 pumping at Buckman well field on the Rio Tesuque and Pojoaque River

In a memorandum dated December 14, 1987, from Michael P. Spinks (then of the Hydrology Section of the Technical Division of the State Engineer Office) to M.B. Compton (then chief of the Water Rights Division of the State Engineer Office), a recommendation was made that the McAda-Wasiolek model of the Santa Fe area be adopted "to calculate effects of withdrawals from the Buckman wellfield on the Rio Grande, Pojoaque River and Rio Tesuque." Therein was also a comparison of the McAda-Wasiolek model with an earlier model (the triangular model), based on the Glover-Balmer analytic solution and image well theory, which was utilized by the State Engineer Office for administrative purposes in the Santa Fe area.

By a letter, dated January 14, 1988, from M.B. Compton to V. Phillip Soice (then manager of the Sangre de Cristo Water Company) a method was described whereby the State Engineer Office would calculate effects of withdrawals from the Buckman well field on the Rio Grande, Pojoaque River and Rio Tesuque by summing the residual effects estimated with the triangular model of pumping from 1972-1987 with pumping effects estimated with the McAda-Wasiolek model from 1988 forward. That method was agreed to and adopted only for the Buckman well field. The triangular model was utilized for the administration of all other water rights in the Santa Fe area until 1996.

It now appears that during the process of formulating the recommendation and the issuance of the subsequent letter some breakdown in communication or expectation occurred between Mr. Spinks and Mr. Compton. My reason for postulating such a breakdown is that the tables of residual effects provided by Mr. Spinks ended in the year 2010. At that point in time the residual effects on the Rio Grande had gone to zero but the effects on the Pojoaque River and Rio Tesuque had not. Re-running the triangular model to provide estimates of residuals on those tributaries would not have been particularly onerous at the time. Unfortunately, the machine on which the triangular model was run is inoperative and the program methodology has now been lost.

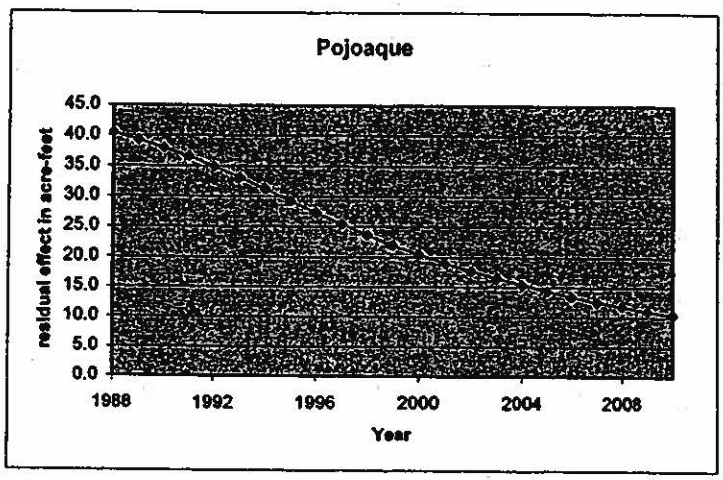
I interpret Mr. Spink's memorandum to imply that he felt that the residual peak value was to be used within the method described in Mr. Compton's letter. That practice has not been followed. Rather, the declining residuals in the Spink's tables have been added to the rising pumping effects estimated using the McAda-Wasiolek model. At the present time, a pending application by the City of Santa Fe has provoked the question of how to forecast the non-zero residual effects on the Pojoaque River and Rio Tesuque to a point in the future where they would go to zero and what values to add onto the McAda-Wasiolek estimates at times past the year 2010.

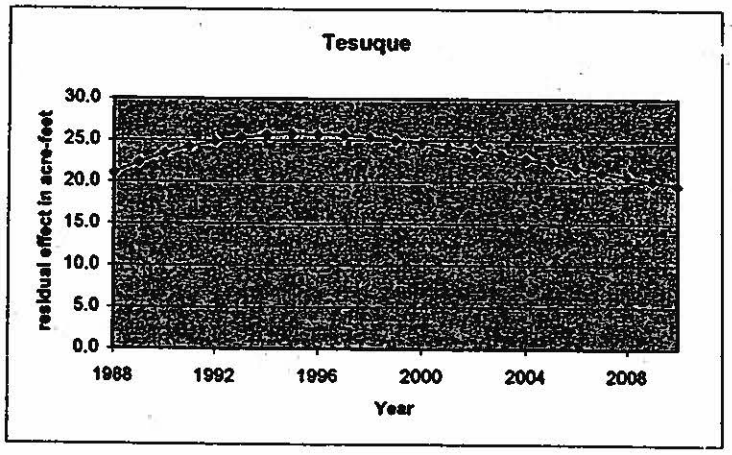
To answer that question I have prepared a spreadsheet and graphs of the Spink's tables using Microsoft Excel and regressed best-fit linear functions to the data for each tributary. The Pojoaque River data allowed for a good fit using the entire 1988-2010 data set. The Rio Tesuque data was more problematic because the data peaked and changed slope within the relevant time period. Therefore, I iteratively experimented with the data until I achieved a good fit using the 1998-2010 data set. The PROJECTION.XLS spreadsheet with the original graphs and projected values is attached. The graphs illustrating the best-fit lines with the linear functions and root mean squared error values are also attached.

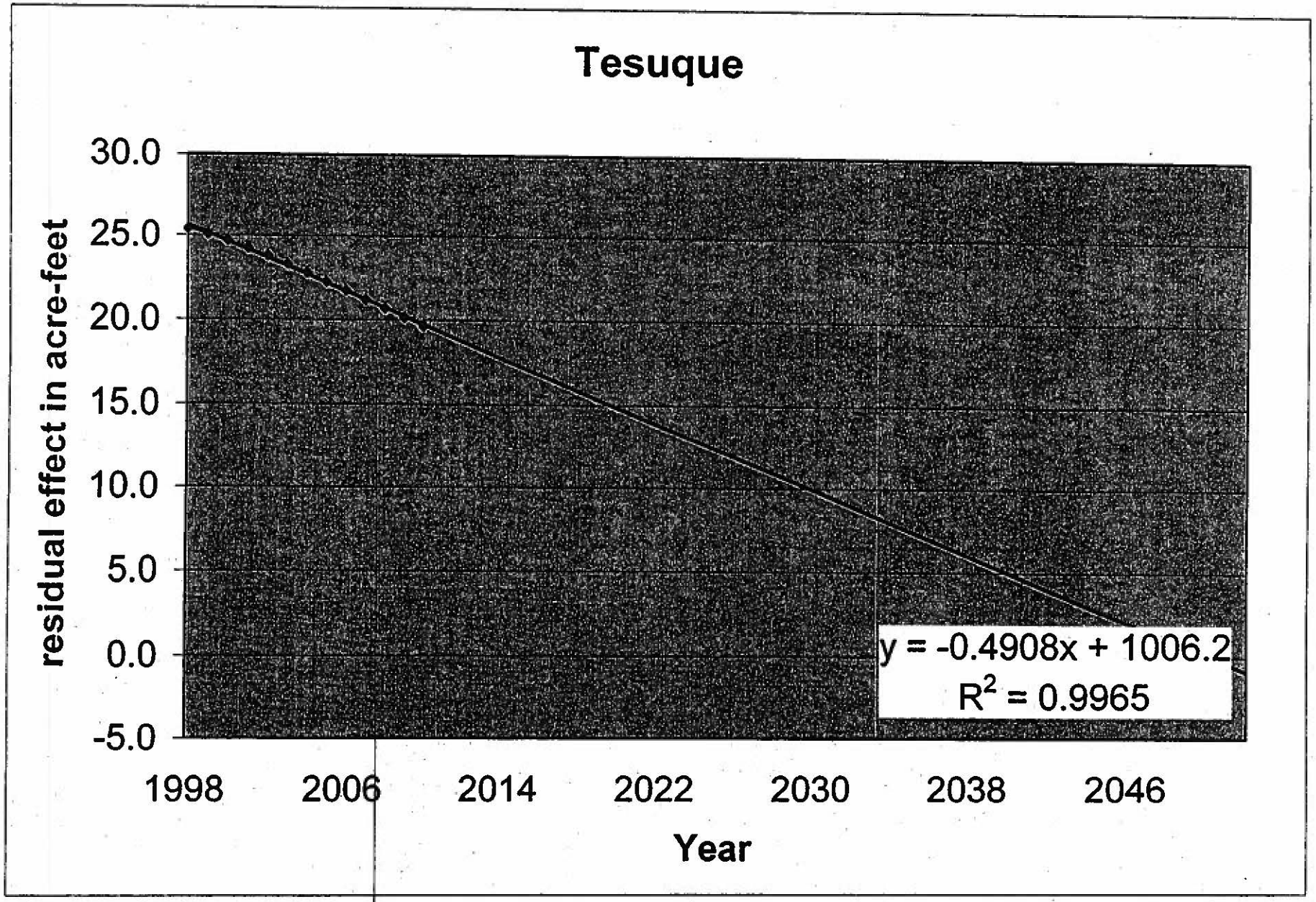
I feel confident in recommending the use of the forecast values for the estimation of effects to tributary surface sources in the present and subsequent calculations concerning the Buckman well field.

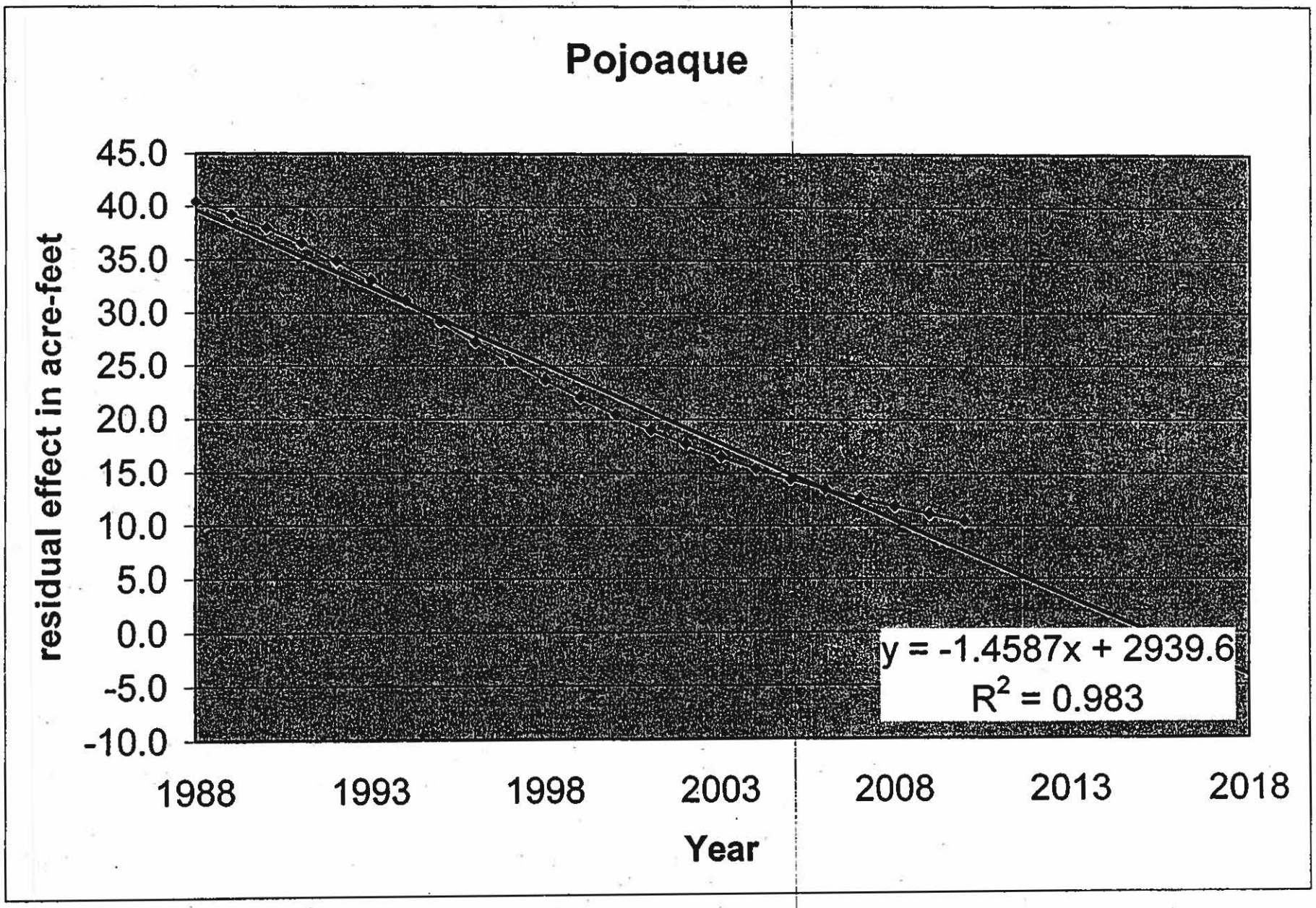
projection.xls

Year	#	Rio Pojoaque-Nambe Residual Impact of 1972-87 Pumping (acre-feet) Analytical Model	Year	Rio Tesuque Residual Impact of 1972-87 Pumping (acre-feet) Analytical Model
1988	1	40.432	1988	21.015
1989	2	39.244	1989	22.333
1990	3	37.971	1990	23.391
1991	4	36.557	1991	24.227
1992	5	34.928	1992	24.868
1993	6	33.112	1993	25.327
1994	7	31.185	1994	25.615
1995	8	29.226	1995	25.747
1996	9	27.296	1996	25.737
1997	10	25.439	1997	25.608
1998	11	23.678	1998	25.378
1999	12	22.028	1999	25.067
2000	13	20.491	2000	24.691
2001	14	19.068	2001	24.265
2002	15	17.753	2002	23.800
2003	16	16.543	2003	23.308
2004	17	15.429	2004	22.797
2005	18	14.404	2005	22.273
2006	19	13.462	2006	21.743
2007	20	12.595	2007	21.212
2008	21	11.797	2008	20.683
2009	22	11.061	2009	20.157
2010	23	10.383	2010	19.639
2011	24	6.151	2011	19.258
2012	25	4.693	2012	18.767
2013	26	3.234	2013	18.276
2014	27	1.775	2014	17.785
2015	28	0.316	2015	17.295
2016	29	-1.142	2016	16.804
2017	30		2017	16.313
2018	31		2018	15.822
2019	32		2019	15.331
2020	33		2020	14.841
2021	34		2021	14.350
2022	35		2022	13.859
2023	36		2023	13.368
2024	37		2024	12.877
2025	38		2025	12.387
2026	39		2026	11.896
2027	40		2027	11.405
2028	41		2028	10.914
2029	42		2029	10.424
2030	43		2030	9.933
2031	44		2031	9.442
2032	45		2032	8.951
2033	46		2033	8.460
2034	47		2034	7.970
2035	48		2035	7.479
2036	49		2036	6.988
2037	50		2037	6.497
2038	51		2038	6.006
2039	52		2039	5.516
2040	53		2040	5.025
2041	54		2041	4.534
2042	55		2042	4.043
2043	56		2043	3.552
2044	57		2044	3.062
2045	58		2045	2.571
2046	59		2046	2.080
2047	60		2047	1.589
2048	61		2048	1.098
2049	62		2049	0.608
2050	63		2050	0.117
2051	64		2051	-0.374






Tesuque
residual effect in acre-feet
30.0
25.0
20.0
15.0
10.0
5.0
0.0
-5.0
1998
2006
2014
2022
2030
2038
2046
Year
y = -0.4908x + 1006.2
R2 = 0.9965


Pojoaque
residual effect in acre-feet
45.0
40.0
35.0
30.0
25.0
20.0
15.0
10.0
5.0
0.0
-5.0
-10.0
1988
1993
1998
2003
2008
2013
2018
Year
y = -1.4587x + 2939.6
R2 = 0.983